

Latent Representation Probing for Toxic Comment Detection

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01 Why Toxicity Detection Needs Context

The Scale Problem

- Millions of comments daily make manual moderation impossible.
- Standard keyword filters miss subtle semantic context and sarcasm.
- Fine-tuned classifiers require significant compute for deployment.

The Efficiency Gap

- Full LLM inference is computationally prohibitive at scale.
- High latency creates bottlenecks in real-time user interactions.
- Need for lightweight, contextual, and accurate detection.

02 Technical Approach: Probe the Latent Space



We extract internal activation vectors from specific transformer layers of a frozen base model (e.g., Gemma). A lightweight probe is trained to distinguish toxic from benign text based on these high-level semantic features.

03 Dataset and Feature Extraction

Source & Splits

- Public Toxicity Benchmarks (Civil Comments / Jigsaw)
- Train / Test Split: 80% / 20% PENDING NOTEBOOK
- Binary Labels: 0 (Benign) vs 1 (Toxic)

Extraction Pipeline

- Base Model: Gemma-2b / 7b PENDING NOTEBOOK
- Layer Selection: Intermediate Transformer Layers
- Pooling: Mean-pooling / Last-token activations
- Library: TransformerLens / Hugging Face

04

Evaluation Framework

Experimental Results

METRIC	BASELINE (KEYWORD)	PROBING CLASSIFIER
Accuracy	PENDING	PENDING
Precision	PENDING	PENDING
Recall	PENDING	PENDING
F1-Score	PENDING	PENDING
ROC-AUC	PENDING	PENDING

Values to be populated from the final notebook execution. The evaluation focuses on the performance gain of latent probing over traditional rule-based baselines.

05 Impact, Risk, and Responsible Use

Safety Impact

Community Safety: Real-time detection to filter harmful content.

Moderator Wellbeing: Reducing mental burden via automated screening.

Operational Efficiency

Low Latency: Target < 50ms classification per post.

Energy Savings: Avoiding full LLM fine-tuning and inference.

Risk Mitigation

False Positives: Calibrated thresholds and human-in-the-loop review.

Bias: Continuous evaluation on adversarial and OOD test sets.

Conclusion

Latent representation probing offers a lightweight, contextual, and efficient alternative for toxic comment detection at scale.

Next: Multi-label Toxicity & Cross-lingual Probing